CYBERSECURITY RISK ASSESSMENT & MITIGATION

BLUE HARBOR SOLUTIONS | SIMULATED GRC CASE STUDY

Portfolio assessment • Graduate-level Cybersecurity Risk Management coursework • Old Dominion University

Executive Summary

This report documents a scenario-based cybersecurity risk assessment for Blue Harbor Solutions, a fictional organization operating in a hybrid and remote-work environment. The assessment identifies sensitive information assets, evaluates threats and vulnerabilities, scores three risks using likelihood and impact, and recommends appropriate risk treatment strategies. Phishing and credential theft and insecure remote access each received a score of 9; use of personal devices received a score of 6.

1. Scope and Critical Assets

The scenario includes financial and employee records, a cloud-based customer database, and employee email accounts. These assets support business operations and may contain sensitive organizational, employee, and customer information.

2. Threats and Vulnerabilities

Malware and unpatched devices: Missing security patches create opportunities for exploitation of company devices and systems.

Phishing and credential theft: Insufficient current phishing-awareness training contributes to successful credential compromise.

Unauthorized access: Inconsistent multifactor authentication increases exposure when employee credentials are stolen.

3. Risk Assessment Methodology

The academic assessment used a three-point likelihood and impact scale: 1 = Low, 2 = Moderate, and 3 = High. **Risk score = likelihood × impact.** The qualitative ratings below are those assigned in the original lab; they are not statistically derived probabilities or monetary loss estimates. The original lab labels both scores of 6 and 9 as High Risk.

Risk	Likelihood	Impact	Score	Rating
Personal devices for work	2 – Moderate	3 – High	6	High
Insecure remote access	3 – High	3 – High	9	High
Phishing / credential theft	3 – High	3 – High	9	High

4. Risk Prioritization

Phishing and credential theft was prioritized because the scenario describes employee credentials already being compromised and used to access organizational systems, including customer information. Insecure remote access also scored 9 and warrants prompt attention because employees connect through home and public Wi-Fi networks. These risks share the highest assessment score, but the reported credential compromise provides a specific basis for prioritizing phishing-related response actions.

5. Risk Treatment and Recommendations

Phishing and credential theft — Mitigate: Secure compromised accounts, investigate unauthorized activity, enforce multifactor authentication, deliver phishing-awareness training, and establish a formal suspicious-email reporting process.

Use of personal devices — Avoid: Prohibit unmanaged personal devices for company work and provide approved, company-managed devices with consistent security settings and patching.

Insecure remote access — Mitigate: Require a company-managed VPN, multifactor authentication, endpoint protection, and regular security patching for remote connections.

6. Suggested Validation Activities (Portfolio Extension)

To support future audit or GRC review, an assessor could request MFA configuration exports, phishing-training records, incident reports, approved-device inventory, endpoint patch reports, and remote-access configuration evidence. These are proposed verification activities only; no technical control testing or independent audit was performed for this academic scenario.

7. Conclusion

Blue Harbor Solutions should prioritize credential compromise response and stronger authentication while addressing insecure remote access and unmanaged personal devices. Applying differentiated risk treatments—mitigation for phishing and remote access, and avoidance for unmanaged personal-device use—provides a documented basis for management action. Residual risk should be reassessed after safeguards are implemented and verified.

Source and Limitations

Source: *Case Study Lab 1*, user-provided completed academic assessment. This report reformats and summarizes the submitted analysis for a public-facing professional portfolio. Blue Harbor Solutions is fictional; findings, risk scores, and recommendations are scenario-based, not observations from a real organization. No control effectiveness or remediation completion is claimed.